

COVID-19 Detection from Chest X-Ray Images Using GBM with Comparative Analysis

Abisek Dahal(✉), Abu Motaleb Rony, and Soumen Moulik

Department of CSE, National Institute of Technology, Shillong, Meghalaya, India
abiseknitskm@gmail.com

Abstract. Effective disease management and control depend on the quick and precise diagnosis of COVID-19. Chest X-ray(CXR) imaging has gained prominence as a convenient and accessible diagnostic tool. This study presents a Gradient Boosting Machines(GBM) and Convolutional Neural Networks (CNNs)-based model of COVID-19 detection, along with a comprehensive comparative analysis of the performances of various other machine learning (ML) algorithms. We compiled collected and pre-processed datasets comprising CXR of healthy people and those with COVID-19 positive. The model was trained with diverse CNN architectures and hyperparameter tuning optimized their performance. Algorithm effectiveness was assessed using evaluation metrics including accuracy, precision, recall and F1-score. Experimental results revealed the strengths and weaknesses of each algorithm in COVID-19 detection and the benefit of using these algorithms. The CNN and GBM model successfully differentiate between COVID-19-positive cases and healthy people with a remarkable accuracy rate of 97.4% but GBM take less computation time as compared to CNN.

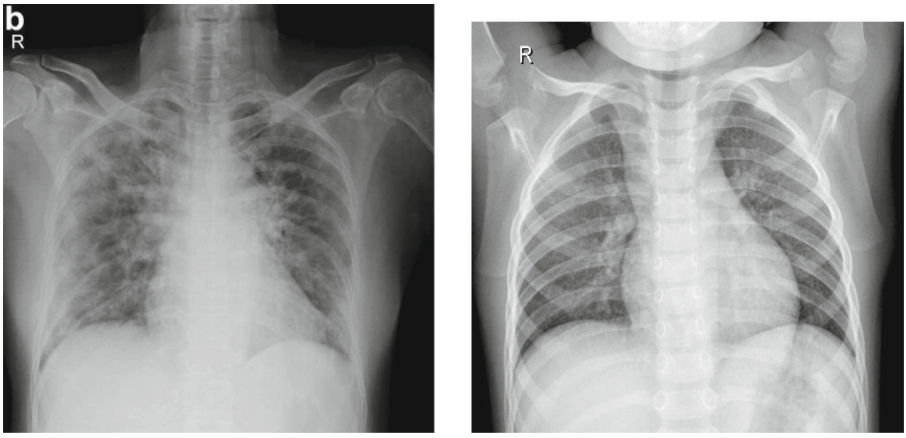
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1 Introduction

Effective disease management and control depend heavily on the timely and precise diagnosis of diseases. CXR imaging has become one of the more practical and accessible diagnostic methods in many cases. We give a thorough comparison analysis of COVID-19 detection using machine learning algorithms with CXR in this paper. To carry out this study we meticulously collected and preprocessed datasets comprising CXR images from COVID-19-positive patients as well as healthy individuals. Ten machine learning algorithms were subsequently put into practise, including Support Vector Machines (SVM), K-Nearest Neighbours (K-NN), AdaBoost, Decision Tree, Gradient Boosting Machine, Logistic Regression, Naive Bayes, Random Forest, Recurrent Neural Network (RNN) and XG Boost (eXtreme Gradient Boosting). These algorithms were trained using

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various architectures and their performance was optimized through hyperparameter tuning. To evaluate the effectiveness of the algorithms a range of evaluation metrics such as accuracy, precision, recall and F1 score were employed. The experimental results of our study provided insights into the strengths and weaknesses of each algorithm in the detection of COVID-19. Through a comparative analysis against benchmark models, we gained a deeper understanding of their relative performance.



(a) with COVID-19 (b) without COVID-19

Fig. 1. Chest X-ray of an individuals

The Fig. 1a presents a CXR of an individual infected with COVID-19. This image exhibits specific characteristics associated with the disease, such as ground-glass opacities, bilateral infiltrates and consolidation. These radiographic findings are often observed in COVID-19 patients and indicate the presence of pneumonia caused by the virus. The abnormal opacities can vary in size, shape and distribution reflecting the severity and stage of the infection. In contrast Fig. 1b displays a CXR image of a patient without COVID-19.

A comparative analysis of this image with the COVID-19-infected CXR provides a visual contrast that allows us to identify the distinct features characteristic of COVID-19. Furthermore we employed visual interpretation techniques, including visualization and analysis to shed light on the decision-making processes of the models. The results of our study show that the CNN and GBM algorithm performed better than the other algorithms that were compared, displaying great precision and specificity. This shows that it may have the ability to detect COVID-19 accurately from CXR images. The GBM and CNN model successfully distinguished between COVID-19-positive cases and healthy people with a remarkable accuracy rate of 97.41%.

2 Related Works

Several studies have been conducted to evaluate the performance of machine learning algorithms for COVID-19 detection from CXR images. One study proposed an automatic deep learning(DL) classification method using CNN [1]. The authors in [2] proposed a amalgamation of deep learning and transfer learning. They came to the conclusion that the accuracy could be improved by two image pre-processing processes and the creation of pseudo-color images. To investigate the possibility of a DL decision-tree based model was proposed in [3].

The author in [4] proposed an automated system for COVID-19 using a DL algorithm using limited training data sets. They achieved an accuracy of 91.3% and demonstrated the potential of DL for COVID-19 diagnosis. An intelligent deep CNN-based COVID-19 detection algorithm was proposed in [5] which achieved an accuracy of 96.5% and demonstrated the potential of it for COVID-19 diagnosis. They used a two-step clustering techniques incorporating bootstrap aggregation with a number of NN methodologies to improve diagnostic sensitivity and lower error rates.

The author in [6] introduce classification framework called SLIPR that tackles the problem of inaccurate labels which can hinder the performance classification. Moreover to replicate a noisy environment they just use random noise. The author in [7] proposed a deep-learning method that achieved an accuracy of 96.5%. A systematic review of machine learning to ascertain and predict for COVID-19 using CXR as well as CT scans was conducted [8]. X-rays can act as an essential visual assistance and diagnostics framework for visualization tests and illness diagnosis in clinical process [10]. The author in [12] presents a two-phase approach for real-time COVID-19 detection using CXR combining a D-CNN and ELMs for classification. The author in [13] proposed a 'DeepChest' to learn desirable features from images. In comparison to earlier methods, "DeepChest" requires a far smaller number of max-pooling layers, fewer convolutional layers and fewer training iterations to execute. As a result, its performance overall and capacity for forecasting will suffer.

3 Proposed Model

3.1 Data Collection

Data collection involved the compilation of a diverse dataset consisting of CXR images from both positive and negative cases. Standard reprocessing techniques namely resizing, normalization and noise reduction is used to enhance the quality of the dataset. The dataset underwent rigorous review ensuring precise labeling with bounding boxes or segmentation masks. We have used CXR images image to evaluate the proposed architecture. Due to the high sensitivity of medical data, we have utilized a combination of publicly available dataset sourced from Kaggle (https://www.kaggle.com/datasets/tawsifurrahman/covid19-radiography-database) to conduct this research. To ensure transparency and replicability, we followed an 80% training and 20%

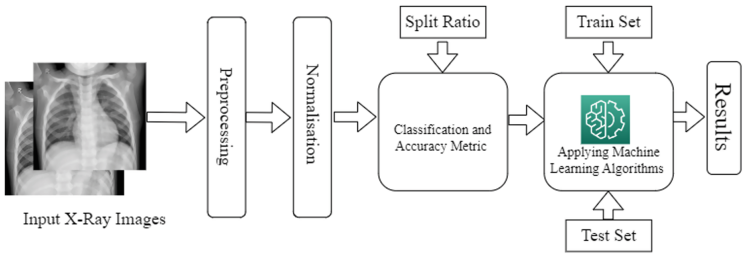


Fig. 2. Architecture of Proposed System

testing data split which will ensures extensive learning from a large portion of data. Followed by an unbiased evaluation of model performance on unseen data balancing effective training with accurate validation. The dataset encompasses 2905 chest X-ray images, encompassing 219 COVID-19 cases, 1341 normal cases and 1345 viral pneumonia cases.

3.2 Preprocessing

The resultant dataset comprises of X-ray images with diverse shapes and resolutions. To standardize the input dataset and ensure consistency throughout, we resized all images to a fixed dimension of 224×224 pixels. This standardization is essential for training Machine Learning models.

3.3 Normalisation

Furthermore, we implemented data augmentation techniques to augment the number and diversity of chest scans. Utilizing a large-scale dataset is crucial for training respective machine learning models and achieving satisfactory results. By employing data augmentation, we introduced variations to the initial images, including rotations, zooms, horizontal flips and shifts. Min-Max normalization and Z-score normalization is used to standardize the X-ray images. Min-Max normalization uniformly scales the pixel values while Z-score normalization ensures they conform to a normalized distribution. It is crucial for preparing the images for accurate and consistent analysis by machine learning algorithm. These techniques significantly increased the size and diversity of our dataset contributing to the overall goal of enhancing the model's performance.

3.4 Evaluation Metrics

We evaluate the performance of each algorithm by using normal evaluation technique used to evaluate machine learning model. The *Precision (P)* is calculated by dividing the number of true positives by the sum of true positives (TP) and false positives (FP). Mathematically,

$$P = \frac{\mathrm{TP}}{\mathrm{TP} + \mathrm{FP}} \tag{1}$$

Recall (R) is calculated as ratio of true positives to the sum of true positives and false negatives (FN), quantifying the ability of a model to correctly identify positive instances. Mathematically,

$$R = \frac{\mathrm{TP}}{\mathrm{TP} + \mathrm{FN}} \tag{2}$$

A statistic called Mean Average Precision (mAP) is used to assess how well item detection and recognition systems perform. A thorough evaluation of the model's accuracy in detecting and localising objects in distinct categories is provided by calculating average precision for each individual class and then taking the mean across all provided classes.

$$mAP = \frac{1}{n} \sum_{i=1}^{n} AP_i \tag{3}$$

where n is the number of classes and $AP_i$is an average precision for class i. Average Precision (AP) is a performance metric used in object detection and information retrieval tasks to measure the quality of ranked predictions. Mathematically,

$$AP = \frac{1}{TP} \sum_{TP} IR \tag{4}$$

where TP represents true positives at each interpolated recall (IR) level.

F1-score (F1) is an evaluation metric which uses both precision as well as recall and calculated as harmonic mean of the two, providing a balanced measure of a model's accuracy. Mathematically,

$$F1 = \frac{2 \times \mathrm{P} \times \mathrm{R}}{\mathrm{P} + \mathrm{R}} \tag{5}$$

Sensitivity (Sen) is an unit that measures a machine learning model's ability to accurately detect positive instances. It represents the proportion of true positives that the model correctly identifies.

$$Sen = \frac{TP}{TP + FP} \tag{6}$$

Specificity (S) is a measure of a machine learning model's ability to precisely identify negative instances. It indicates the proportion of true negatives that the model accurately recognizes. Mathematically,

$$S = \frac{TN}{TN + FP} \tag{7}$$

Accuracy (A) is the ratio of correctly classified class to the total number of class, reflecting the overall correctness of a model's predictions. Mathematically,

$$A = \frac{TP + TN}{TP + TN + FP + FN} \tag{8}$$

3.5 Algorithms Used in Experiment

To conduct this experiment, we applied machine learning algorithms on proposed architecture in Fig. 2 both the original and normalised dataset. Our objective is to determine the algorithm that achieves the highest accuracy in detecting Covid-19 cases.

The algorithms used in our experiment are as follows:

1. CNN - Convolutional Neural Network
2. LR - Logistic Regression
3. DT- Decision Tree
4. NB - Naive Bayes
5. RF - Random Forest
6. K-NN - K-Nearest Neighbors
7. RNN - Recurrent Neural Network
8. AdaBoost
9. XGBoost
10. GBM - Gradient Boosting Machine
11. SVM - Support Vector Machine

We will compare the performance of these algorithms based on their accuracy.

Table 1. Comparison of Algorithms

Algorithm	P	R	F1	S	A
CNN	1	0.375	0.545	1	0.974
SVM	0.5	0.625	0.555	0.972	0.958
K-NN	0.8	0.5	0.615	0.994	0.974
Decision Tree	0.166	0.375	0.23	0.918	0.896
Naive Bayes	0.161	0.625	0.256	0.859	0.849
Logistic Regressi on	0.444	0.5	0.47	0.972	0.953
Random Forest	1	0.25	0.4	1	0.968
RNN	0.98	0.378	0.6	1	0.958
AdaBoot	0.5	0.25	0.333	0.989	0.958
XGBoost	1	0.25	0.4	1	0.968
GBM	1	0.370	0.540	1	0.974

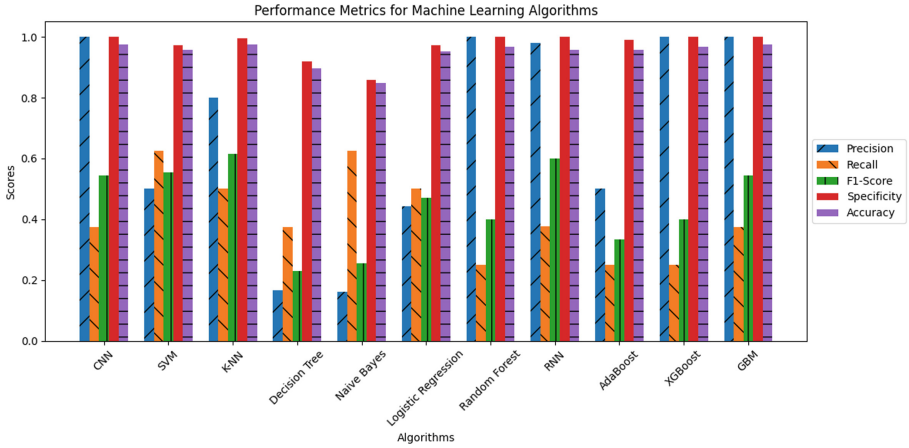


Fig. 3. Performance metric for Machine Learning Algorithms

4 Result and Simulation

The experiment was conducted using TensorFlow a popular framework for machine and deep learning model development and deployment. Additional libraries employed included Keras, SK-Learn and Scipy enhancing the experimentation process. The experiment necessitated a dedicated GPU on a computer server in google colab with the GPU with 15 GB GDDR5 VRAM for computational capabilities to facilitate efficient computations.

From Table 1 and Fig. 3 it is clear that GBM and CNN achieved the highest precision (P) among all the algorithms indicating a low false positive rate. This means that the CNN was able to correctly classify a large proportion of COVID-19 cases. SVM (Support Vector Machine) had the highest recall(R) of 0.625, suggesting a relatively lower false negative rate. SVM performed well in identifying true positive COVID-19 cases, although it had a slightly higher false positive rate compared to CNN. K-NN exhibited a high specificity(S) value 0.994, indicating a low false positive rate. This suggests that K-NN was effective in correctly classifying non-COVID-19 cases. DT, NB, LR, RF, RNN (Recurrent Neural Network), AdaBoost, XGBoost, and showed relatively lower evaluation metrics compared to GBM, CNN, SVM, and K-NN.

However, all demonstrated performance depends on the application's unique requirements. Overall the results suggest that GBM and CNN performed the best among the evaluated algorithms with high precision and specificity indicating its potential for accurate detection of COVID-19 with X-ray. We subsequently assessed the computational durations of all the algorithms listed in Table 2. We used the python time package to compute the training and testing times. The training time refers to the duration required for a particular model or machine learning algorithm to learn from the provided data. Conversely the testing time pertains to the duration needed for evaluating the results obtain by validating

with new data against the trained data for the model. To estimate the total computation time (C_t), we employed the following formula: $C_t^i = T_t^i + T_s^i$, where T_t and T_s denote the time taken for training and testing the dataset respectively. This process ensures accurate measurements of the overall computational time for each classifier.

Table 2. Computation Time(In Sec.)

Algorithm	Training Time	Testing Time	Total Time
CNN	980.25	60.23	1040.48
SVM	870.49	20.95	891.44
KNN	70.85	12.18	83.03
Decision Tree	33.12	12.32	45.44
Naive Bayes	71.161	13.25	84.411
Logistic Regression	55.4	12.31	67.71
Random Forest	43.11	12.2	55.31
RNN	204.94	18.78	223.72
AdaBoot	16.32	7.25	23.57
XGBoost	32.89	8.25	41.14
GBM	420.12	32.12	452.24

5 Conclusion

Experimental results revealed the strengths and weaknesses of each algorithm in COVID-19 detection. Comparative analysis against benchmark models provided insights into relative performance. The GBM and CNN model stood out as the most effective in detecton COVID-19 using X-ray, achieving a remarkable accuracy rate of 97.41%. It displayed high Recall, F1 score and Precision indicating its ability to accurately identify positive cases. The GBM model's superior performance suggests its potential for clinical use in COVID-19 diagnosis due to less computation time. Further research and validation are necessary to confirm the general suitability and robustness of the GBM model across diverse patient populations and healthcare settings. Based on our computational analysis it seems that GBMs have an edge over CNNs when it comes to computational time making them a more convenient choice in certain situations. However, it is important to remember that CNNs offer their own unique advantages over GBMs. CNNs have a simpler architecture and training process which can make them more approachable and easier to work with. On the other hand, GBMs are an interesting technique that combines multiple weak models, usually decision trees, to create a more robust model. Ultimately, the decision between GBMs and CNNs depends on the specific needs and characteristics of the task you are

tackling. It is crucial to take into account additional elements like computing complexity, scalability and interpretability when choosing an efficient algorithm for practical implementation. In future research we will focus on refining and validating these models for real-world clinical use.

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